Caden Otis

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Professional Summary

An experienced Embedded Software Engineer with a passion for building reliable firmware and systems. Open to opportunities that offer meaningful technical growth in the embedded space.

Education

l**owa State University**, Ames, IA | Graduated May 2025
Bachelor of Science in Electrical Engineering, Minor in Computer Science| GPA: 3.86

Employment Experience

Rice Lake Weighing Systems, Rice Lake, WI June 2025 - Present
Embedded Software Engineer

- Utilize C++ and C to develop software for weighing indicators, creating new features and fixing both major/minor bugs
- Design and build frontend UIs embedded system using HTML, CSS/Less, and JavaScript
- Write and maintain comprehensive unit tests alongside firmware development to ensure system reliability
- Debug low-level communication using logic analyzers across SPI, RS-232, and RS-485 serial interfaces
- Worked on integrating digital load cells and custom communication protocol

John Deere, Ames, IA Sep 2022 – May 2025
Part Time Student - Product Engineering Parallel Coop

- Developed embedded software for features using C++, Object Oriented programming, and Qt Creator that enhance the UI/UX experience on Gen4/5 in-cab displays (improving field boundary adjustments and diagnostics, modifying parsers for CAN messages, and resolving critical data miscalculations, UI element spacing, and text issues).
- Conducted unit testing (GTest), Jenkins automation, and UX tests using Python to ensure software reliability.
- Worked on an Agile Scrum team, performing code reviews and participating in daily standup meetings.
- Utilized a Linux environment for development, version control for multiple repositories, and running simulations.

Projects

Single-Cycle Processor Nov 2024 – Dec 2024
ISU CPRE 3810 – Computer Organization and Assembly Level Programming

- Designed and implemented a processor that can fetch, decode, and execute MIPS assembly instructions, with each instruction being executed per clock cycle.
- Utilized VHDL to implement the digital logic architecture of the processor and QuestaSim to simulate its behavior.

Roomba Obstacle Course Aug 2023 – Dec 2023
ISU CPRE 2880 – Embedded Systems

- Tasked with modifying a Roomba to navigate an obstacle course without hitting obstacles or leaving the course, with the driver not being able to see the robot relying only on sensor data.
- Programmed a TM4C123GH6PM microcontroller using embedded C to control a Roomba via keyboard inputs. Worked with the UART, ADC, and PWM.

Honors and Awards

Kiewit Scholars Program Aug 2022 – May 2025
Iowa State Honors Program Jan 2022 – Aug 2024
Iowa State Dean's List Dec 2021 – May 2025
Iowa State First-Year Honors Program May 2021 - May 2022

Skills

Java | Python | C | C++ | HTML | JavaScript | CSS/Less | Git | GitHub | Linux | Android Studio | MATLAB | Simulink | SPI